

# Spending the Budget: Glide Projection and the Choice Between Deep and Broad Drawing

*Draft A — full manuscript, assembled. A design-research paper in the author's first-person voice. Sections 1–2 establish the principle and its footing; sections 3–6 carry the instrument, the correction, the plates, and the closing thesis. All quantitative claims are verified in the companion addendum A13 ( `check_a13_budget.js` ). The five computer-vision citations were verified against source in July 2026.*

---

## Front matter — the footing this paper rests on

The claim of this paper is not a new geometry. It is a design principle, a drawing method, and — the part the record now settles — a matter of priority and completion rather than invention. Three facts govern the citation posture, and the paper states them early:

1. **The canon, 1979.** In *Graphics for Architecture* (Forseth, with Vaughan; Wiley, 1979), I set down the constructive account of the existing families — orthographic plan, elevation and section; the paraline/axonometric obliques; perspective; and cast shadows — all built from descriptive geometry: fold lines, auxiliary views, cutting planes, measuring lines. That book is the evidence of standing: the work of someone who knew the whole projective canon from the inside, as constructions.
2. **The fourth projection, 1984 — now read to its last plate.** In *Glide Projection: Lateral Architectural Drawing* (Forseth; Van Nostrand Reinhold, 1984), I constructed the projection the canon lacked — glide — a complete, internally consistent measured system, and I distinguished it in the same book from its optically real twin, which I named double-slit projection and set aside because its receding lines curve (pp. 9–10). I derived glide's structure synthetically, stated its consistency conditions as a draftsman states them — no gaps or overlaps within or between lateral armatures, and a common infinity plane (p. 98) — recognized it was mechanizable, and deliberately left its analytic transcription undone, saying so on the page (p. 48). The applications chapter (pp. 103–122) then did two further things whose weight only becomes visible now: it named glide's native subjects — the plan, the site plan, the section, views of buildings and landscapes *in low spatial relief* (p. 103) — and it staged, plate against plate, the comparison between glide and perspective on the same subjects, ending with an explicit invitation to the reader to compare the visual results. The book poses the question this

paper answers. It never states the rule.

3. **The independent rediscovery, 2002–2015.** Working analytically from optics, computer vision arrived at the same projection glide abstracts — the crossed-slit or X-slit camera (Pajdla 2002; Zomet, Feldman, Peleg & Weinshall 2003), placed in a continuous family of linear cameras by Yu & McMillan (2004) — and wrote the closed-form depth law (Yang, Lin, Kang & Yu 2015). None of it knew the architectural system existed. And the anticipation runs deeper than the projection itself: the 1984 book specifies that its freeform glides are not to be taken in from a single station but observed part by part, in a continuous gliding motion, with local believability everywhere and no single position for the whole (p. 100) — which is the viewing protocol of the multiperspective mosaic, stated as a drawing instruction nineteen years before the mosaic was computed (Zomet et al. 2003; Román et al. 2004). It is also the origin of the name.

So the seam is exact, and it is the paper's spine: the seeing was synthetic and mine, in 1979–84; the calculation was analytic and theirs, two decades later; the machine that runs both is the present tense that reunites them. I do not claim the algebra. I claim the construction that contains it, the recognition that it was computable, the design principle it yields — and, now, the completion. The vision literature is cited generously as the independent rediscoverer, not as a source. The one word that carries the novelty is *measured*: glide is a drawing you can dimension, and it takes its seat with the plan and the section, not with the picture.

---

---

## 1. The bargain and the principle

Every architectural drawing strikes a bargain with the third dimension, and every convention we have inherited strikes it differently. The plan and the section keep measure and surrender the view; the axonometric keeps measure and parallelism and surrenders the way a room meets the eye; perspective keeps the eye and surrenders measure — and, the part we tend not to say aloud, surrenders it unevenly, so that a wide façade forced into a single perspective splays and bloats at its edges while a deep nave sits perfectly. We choose among these less by principle than by habit and by what the software defaults to. My purpose here is to show that the choice is in fact governed by a principle, that the principle can be stated precisely, and that it points to a projection tuned for exactly the scenes at which perspective is worst — a projection that, unlike perspective, hands the architect a measured drawing of those scenes.

The principle is a budget. A projection has a fixed quantity of fidelity to spend, and it must

spend it on one axis at the expense of another. Perspective spends lavishly on depth: its cue for distance is diminishing size, a signal of unbounded range, which is why it renders a corridor receding to infinity without complaint — but it pays on breadth, where a wide field forces the picture plane to meet the cone of rays obliquely and the margins stretch, a distortion that is radial, position-dependent, and removable by no single correction. The complementary projection makes the opposite bargain: it spends on breadth, laying wide scenes down gently, and pays on depth, where its distance cue saturates and — in its raw optical form — its receding lines bow. The drawable member of that bargain is the one that refuses the bow: a straight-line idealization, bought at the price of confining itself to the shallow scenes where the bow would anyway be negligible. The two projections are not rivals. They are two settings of one instrument, and the scene in front of you sets the dial.

I depart here from the way such a proposal is usually made, because the complementary projection is not one I am putting forward for the first time. In 1979, in *Graphics for Architecture*, I wrote the constructive account of the projective canon as it then stood. Five years later, in *Glide Projection*, I constructed the member that canon lacked — I called it glide — the straight-line measured system one obtains by keeping the lateral, two-vanishing character of the true optical projection while refusing its curvature, so that it can be drawn with a T-square and triangle. In the same book I named that optical projection, double-slit, and set it aside precisely because its lines curve, which in 1984, at a drafting table, made it unbuildable and unteachable. I knew glide was internally consistent — I stated the closure conditions on the page — and, having just spent a book establishing that these drawings are mechanical constructions in orthogonal, affine, and projective operations, I knew it could be modeled on a computer. I said as much, and I deliberately left the algebra undone. **I built the system and saw that it was mechanizable; I was not the one to mechanize it.** In the two decades that followed, computer vision, reasoning analytically from the optics, rediscovered the double-slit projection glide abstracts, and wrote the closed-form law relating a feature's proportion to its depth — not knowing the drawing system existed. What follows completes the open end I left in 1984: the algebra, the formal footing, and the machine now exist, and with them the design principle can at last be stated with a number and demonstrated live.

So what I claim is a decision and the drawing method that serves it, not a mathematics. The geometry is constructed and cited to its own sources — mine and, for the analytic law, computer vision's. And the decision itself — match the projection's generous axis to the scene's large axis — turns out to have been *posed* in 1984, though never answered. The applications chapter of *Glide Projection* names glide's natural subjects outright — plan, site plan, section, scenes in low spatial relief — sets the same gardens and buildings in perspective, paraline, and glide plate against plate, and closes by inviting the reader to compare the results. The book stages the comparison as an exhibition and stops at the invitation. This paper accepts it: the mechanism, the number, and the correction that turn a

staged comparison into a stated rule. Beneath the rule runs a quieter thesis I will return to at the close: that a thing can be known by construction decades before it is known by formula, and that the geometer who builds it correctly can see it is computable without being the one to compute it.

## 2. The distortion budget

State the two bargains exactly. In perspective the magnification of a feature is inversely proportional to its distance, a factor running the whole interval from unity to zero as the scene recedes; the range is unbounded, so depth resolution is effectively unlimited and a feature's proportions are preserved at every distance — a square stays a square. That last fact is the reason perspective cannot be measured for depth: shape carries no distance information. Breadth is where it pays, and the payment is irredeemable — the marginal distortion of a wide perspective is radial and position-dependent, removable by no global transform, which is why the wide-angle photograph is cropped rather than corrected.

Glide, and the double-slit projection it idealizes, invert every term, and I derived the inversion synthetically in 1984 before I could write it as an equation. Because the governing lines sit at two different depths, the horizontal and vertical scales fall off at different rates; I split perspective's single "unity convergence" into an independent lateral convergence within the surface and a depth convergence into the picture, and proved the two independent by the continuous-vantage-plane argument (*Glide Projection*, pp. 43–44). That two-axis structure is the whole engine: its ratio — the proportion a square casts — is a function of depth. The gift is that proportion now encodes distance, so a glide drawing is a measured drawing, and the 1984 book equipped it twice over. The midpoint method develops a diminishing scale of true depths from the diagonals of a rectangle (pp. 75–76); the scaled glide goes further and lets depth be read *directly with a ruler* along an orthogonal field line, up to a bounding transverse plane that guards the infinity plane, beyond which the diminishing scale resumes (pp. 100–101). Two measuring systems on one sheet — and the boundary between glide and the paraline drawn exactly: if every receding orthogonal is parallel there is no infinity plane and the drawing is a paraline by definition; one convergent family, and the infinity plane, the measured recession, and the glide exist (p. 101). The reference vanishing frame, whose sight line bisects the image into two equal parts, fixes the depth law as a constructed proportion — the 1:2 ratio, stated as a ratio, with the rate of diminution preserved when the ratio of depth interval to image length is held constant (pp. 48–50). The closed-form transcription of that proportion is precisely what I left undone and what the vision literature later supplied (Yang et al. 2015); it makes computable what the construction already guaranteed.

The cost is the mirror of perspective's, and I stated its mechanism in the same book under "eccentric projective effects" (pp. 68–69): only one family of parallels can stay straight, two

of the three axes must lie in transverse planes, and the third — depth — must be orthogonal for straight lines to survive; curvature, twisting, and stretching all grow with depth of field and vanish as the field goes shallow. This is why glide's distance cue is bounded and its native subject shallow. And it is why the correction is a ruler's worth of work rather than an impossibility: at any single depth the distortion is a pure anisotropic scale — a shear-free stretch of one axis against the other — so one global factor squares it for the whole sheet, where perspective's marginal error admits no such repair.

From this the design rule falls out with the force of arithmetic. **Match the projection's benign axis to the scene's large axis.** Deep and narrow — a nave, a light-well, a corridor, a stair section — belongs to perspective, whose unbounded depth cue has room to work and whose breadth penalty never engages. Broad and shallow — a façade, a wide wall section, a site of massing blocks, a floor plan read as a horizontal cut — belongs to glide, whose gentle breadth and correctable anisotropy keep the whole width legible and whose shallow depth is all such a scene asks. The plan, which is nothing but a shallow section through a building, is glide's most natural application of all.

It is here that the paper adds what the record does not yet hold, and the record — now read to its last plate — makes the addition sharper, not smaller. The optical projection and its analytic depth law belong to computer vision; the canon and its unification belong to the classical literature and to the General Linear Camera. The subject class, remarkably, already belongs to 1984: the applications chapter of *Glide Projection* opens by declaring glide's uses — plan, site plan, section, views in low spatial relief — and then walks the Belvedere garden, the Villa Farnese, the Villa Medici, the Villa Lante, the Villa di Papa Giulio, and a Palladian section through perspective, paraline, and glide in matched plates, inviting the reader to judge. What the book does not do — what no one has done, in the vision literature that sought to minimise this distortion in service of a photograph, or in the architectural-representation tradition that knows plan, section, axonometric, and perspective and no fourth measured thing — is state the *principle* the plates enact: that these two projections partition the space of architectural subjects between them; that the partition is governed by the opposition of a bounded depth cue to an unbounded one; that the designer should choose deliberately by a scene's proportions; and that one of the two yields a measured drawing for the broad, shallow subjects perspective serves worst. In 1984 I chose the subjects correctly and exhibited the choice; I did not state the rule that made it correct. That rule, the number behind it, the measured drawing it licenses, and the completion of a system left open on the page — those are the contribution.

### 3. The instrument with a dial

A reader who has followed the budget will want to see the two projections as one adjustable thing rather than two rival diagrams, and they are. Take the perspective station point and

split it — not metaphorically but by construction — into two lines at two depths. Let a single parameter  $D$  open the gap: the vertical governing line recedes to  $Z_v = Z_p - D/2$ , the horizontal to  $Z_h = Z_p + D/2$ , the two straddling the old station point symmetrically. At  $D = 0$  the lines coincide, the two convergences fuse into one, and the drawing is an ordinary one-point perspective. Open  $D$  and the convergences separate: the image acquires its two independent vanishing behaviors, the aspect ratio of a frontal square begins to encode depth, and the drawing becomes a glide. Push  $D$  wider and the depth cue sharpens while the clean band narrows, exactly as the budget identity requires. One knob runs the whole family, pinhole at one end, wide glide at the other, and the office method of the 1984 book — the image of a point found as the intersection of two planes, each through the point and one governing line (*Glide Projection*, pp. 45–59) — is simply this construction performed at a chosen  $D$ .

That the family is continuous, and that these two projections are neighbors on it rather than strangers, is the content the vision literature supplied independently: Yu and McMillan's *General Linear Camera* places pinhole, orthographic, pushbroom, and two-slit cameras in one space of ray-planes, with the crossed-slit — glide's optical original — the immediate term past perspective. The analytic home of the dial is theirs; the dial itself, as a drawing operation an architect performs with a governing line and a straightedge, is the 1984 construction. The paper cites the first and demonstrates the second.

The demonstration is where the reader stops reading and starts turning the knob. The interactive companion carries  $D$  as a live control: drag it from zero and watch a scene pass from perspective through glide, the vanishing structure splitting in real time, the frontal squares stretching as the aspect-ratio cue engages. What was a static plate in 1984 — glide beside perspective, the reader *invited to compare* — becomes a single continuous morph between them, the comparison the book staged now animated along its missing parameter. The boundary conditions are the book's own: slits coincident gives perspective; all receding orthogonals parallel gives the paraline, which by definition has no infinity plane (p. 101); between those limits lies the glide, with its infinity plane and its measured recession. The dial does not invent the family. It makes visible the axis the canon always had and never drew.

## 4. The correction

Glide's cost, established in section 2, is a per-depth anisotropy: at any single working depth the frontal plane maps by a pure affine stretch of one axis against the other, identical at every position across the field. That last clause is the whole of why the cost is payable. A distortion that is the same everywhere on the sheet is undone everywhere by the same operation — and the operation is a single reciprocal scale.

Concretely: read the aspect ratio a known square casts at the working depth — call it the

gauge,  $AR(z_0) = (Z_v/Z_h) \cdot (z_0 - Z_h)/(z_0 - Z_v)$ . Multiply the compressed axis by its reciprocal,  $1/AR(z_0)$ . The sheet is now square at  $z_0$ , exactly, at every field position — verified to machine precision in A13.3, and implemented in the interactive as the  button that squares the gauge with one factor. This is anamorphic correction in the classical sense the book already indexes under *anamorphic* (p. 31), now given its coefficient: not an artist's trial-and-error unsquaring but a computed reciprocal read straight off the depth.

Two honest boundaries frame the operation. First, it is exact only on the reference plane; through a scene of depth extent  $\Delta$  the residual anisotropy after squaring is  $\epsilon_x = D \cdot \Delta / (2(z_0 - Z_v)(z_0 - Z_h))$  to first order (A13.3), growing linearly with the depth of field and vanishing as the field goes shallow. At the rig constants and a 1% tolerance this buys a clean band of about four units per percent — a twenty-unit band at 5% — beyond which one reintroduces the diminishing scale the way the scaled glide reintroduces it past its bounding transverse plane (pp. 100–101). Second, the correction addresses the first-order anisotropy, not the second-order bow of receding lines; but the bow is  $O(\Delta^2)$  (A13.6), subordinate to the anisotropy inside any working band for which the squaring was worth doing. The correction is therefore what the section title promises: a ruler's worth of work — read the gauge, apply one factor — where perspective's marginal error, radial and position-dependent, admits no global repair at all.

## 5. Plates

The plates do one job: show the rule deciding, correctly, on real architectural subjects, and once show it deciding against a subject chosen to earn the wrong answer. Each is a scene rendered through the dial at the  $D$  its proportions call for, with the gauge factor applied, and each carries its  $\Delta \cdot D$  against  $B^2$  in the caption so the reader sees the budget arithmetic land.

*The wide section.* A broad, shallow building section — many bays across, little depth into the sheet — is glide's home ground. Here  $B$  is large and  $\Delta$  small, so  $\Delta \cdot D \ll B^2$  by a wide margin; the whole width stays legible where a single perspective would splay its end bays, and the squared gauge returns true story heights across the span. Cite the 1984 vanishing-axis and double-herringbone glide sections (pp. 111, 113) as the hand-drawn ancestors; the plate is the same subject with the number attached.

*The floor plan with its wall-depth reveal.* A plan is a shallow section through a building, maximal in breadth and minimal in depth — glide's most natural subject by arithmetic, since the ratio  $\Delta \cdot D/B^2$  is smallest of all. The reveal is the payoff the orthographic plan cannot give: because proportion encodes depth, wall thickness and the drop into a window jamb read directly off the sheet as measured recession, not as a separate section. This is the plate that earns the word *measured*.

*The site plan in relief.* Massing blocks on a broad ground plane, low spatial relief — the

book's own stated domain (p. 103, "landscapes in low spatial relief"). The relief reads as glide depth; the correction squares the ground grid; the breadth that would bow a wide perspective lies down gently under glide's nearer-slit convergence.

*The courtyard.* An interior looked into from above — the aerial one-point courtyard of the 1984 applications chapter (pp. 108–109), now rendered as glide so that the four walls' depth and the floor's true proportions coexist on one measured sheet.

*The deliberately wrong case.* Force a deep, narrow subject — a corridor or nave receding far into the sheet — into glide. Now  $\Delta$  is large and  $B$  small, so  $\Delta \cdot D \gg B^2$ ; the budget verdict is perspective, and glide pays its full price. The plate shows it honestly: the aspect-ratio cue saturates toward  $Z_v/Z_h$  and stops resolving depth, the bow the correction cannot reach grows with the depth of field, and the residual  $\varepsilon_x$  runs past any tolerance the single gauge factor can meet. The caption reports the number that predicted the failure before the drawing was made. Showing the budget's cost is not a concession; it is the proof that the rule is a rule and not a preference.

## 6. Construction and computation

The plates close the practical argument; a quieter one remains. This paper has completed a system I left open in 1984 — but the shape of that completion is itself the point, and it is worth stating plainly because it runs against a common assumption about how knowledge in a mathematical field is supposed to arrive.

The assumption is that one knows a geometry when one can compute it: that the formula is the knowledge and the drawing merely its illustration. The history of glide says otherwise. In 1984 I built the projection as a construction — governing lines, intersecting planes, a measured-depth apparatus, closure conditions stated as a draftsman states them — and I knew it was consistent, knew it was mechanizable, and said on the page that its effects could be verified with definitions and theorems I was giving *without proving* (p. 48). That is knowledge by construction: complete, correct, and deliberately stopped short of algebra. The formula came two decades later, from a different discipline reasoning from optics, not knowing the drawing existed — the crossed-slit camera rediscovered (Pajdla 2002; Zomet et al. 2003), placed in its family (Yu & McMillan 2004), and finally written as a closed-form depth-from-aspect-ratio law (Yang et al. 2015). Their abstract opens on the very fact my construction turned on: that in perspective a frontal object's aspect ratio is invariant to depth, and that the crossed-slit breaks this invariance into a usable depth cue. They arrived analytically at what the drawing had guaranteed synthetically.

So the thing was known-by-construction in 1984 and known-by-formula in 2015, and only now, with the algebra and the machine joined, is it known-by-machine — a drawing you can dimension, compute, and turn on a dial. Three books trace the arc: *Graphics for*



*Architecture* drew the projections the canon already had; *Glide Projection* constructed the one it lacked; the present work computes it. The seam between the second and the third is the seam between seeing and calculating, and it is exact — the seeing was mine and synthetic, the calculation theirs and analytic, and neither is diminished by the other.

Two closing witnesses, both from the 1984 book's last pages, seal the thesis rather than merely decorating it. The first is the name. The book instructs that its freeform glides be read not from a single station but part by part, in a continuous sweeping motion, with local believability everywhere and no one position commanding the whole (p. 100) — the exact viewing protocol of the multiperspective mosaic that computer vision would build nineteen years later, and the reason the drawing is called *glide*. The seeing anticipated not only the projection but the way it must be looked at. The second is the reach backward: the applications chapter closes on a Pompeiian wall mural reconstructed as a double-herringbone glide (p. 122), tying the fourth projection to the book's own history of pictorial space (Viola-Zanini 1629; Pozzo 1693; pp. 19–23). Antiquity's scenographers drew, by eye and intuition, what the fourth projection draws by rule — and the rule, now, computes. The geometer who builds a thing correctly can see that it is computable without being the one to compute it. That is the whole claim, and glide is its proof.

---

## Status of the record

The reading of *Glide Projection* is complete through the index (p. 128); p. 126 is blank, pp. 123–125 are three further application plates (freeform glide plaza, Villa Medici perspective, split axonometric), and pp. 127–128 are the index. The applications chapter contains no deferred algebra and no quantitative rule; the p. 48 statement — demonstration “without proving” — holds to the last page. What it contains instead is the named subject class (p. 103), the second measuring apparatus (pp. 100–101), the staged glide-versus-perspective comparison with its explicit invitation to the reader (pp. 103–125), the gliding-view protocol (p. 100), and the Pompeiian closure (p. 122). The book's own index carries direct entries for *double-slit projection* (pp. 9–10), *anamorphic projection* (p. 31), *vantage plane* (pp. 49–56, 65), *eccentric projective effects* (pp. 68–69), and the *staging transversal/orthogonal* apparatus (pp. 75, 55, 89) — so the priority terms can be cited as the book indexes them, foreclosing any “later reinterpretation” objection. The open thread of v2 is closed, and the claim paragraph above has been recalibrated from “the decision appears never to have been made” to the stronger and more accurate position: *the decision was staged and its subjects were chosen correctly in 1984; the principle governing it is stated here for the first time*.

The five computer-vision citations have been verified against source (title, authors, venue, pagination, and — for the two load-bearing ones, Zomet 2003 and Yang 2015 — abstract

content). No substantive correction was needed; the reference list below carries the resulting bibliographic refinements. Yang 2015's abstract states, in the vision literature's own words, both the perspective aspect-ratio invariance the paper opens with and the depth-dependent-aspect-ratio law it attributes to them; Zomet 2003 independently records that a pinhole is the special crossed-slit whose slits intersect, corroborating the paper's "reduces to perspective when the two lines coincide."

---

## Works to cite

### The author's own, bracketing the argument:

- Forseth, K., with Vaughan, D. (1979). *Graphics for Architecture*. New York: Wiley. [2nd ed., 1997.] — the constructive account of the existing canon; establishes standing.
- Forseth, K. (1984). *Glide Projection: Lateral Architectural Drawing*. New York: Van Nostrand Reinhold. — the fourth projection constructed; double-slit named and distinguished; the algebra deliberately deferred; the subject class named and the comparison staged. (See pp. 9–10, 43–50, 68–69, 75–76; closure conditions p. 98; gliding view p. 100; scaled glide and the paraline boundary pp. 100–101; applications and the invitation to compare pp. 103–122; Pompeiian mural p. 122.)
- [the present work] — the algebra, the double-number formalism, the duality, and the interactive machine that complete the 1984 construction.

### The independent rediscovery (cited as such, not as source). All five verified against source, July 2026:

- Pajdla, T. (2002). *Geometry of Two-Slit Camera*. Research Report CTU-CMP-2002-02 (March 2002), Center for Machine Perception, Czech Technical University, Prague. [Peer-reviewed twin, if a journal citation is preferred: Pajdla, T. (2002). Stereo with oblique cameras. *International Journal of Computer Vision* 47(1–3), 161–170.]
- Zomet, A., Feldman, D., Peleg, S., & Weinshall, D. (2003). Mosaicing new views: the crossed-slits projection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 25(6), 741–754. DOI 10.1109/TPAMI.2003.1201823. — the projection rediscovered, and the mosaic whose viewing protocol the 1984 book had specified; also the source of the pinhole-as-intersecting-slits limit.
- Yu, J., & McMillan, L. (2004). General linear cameras. In *Computer Vision — ECCV 2004*, LNCS vol. 3022, pp. 1–14. Springer. — the unification of pinhole, orthographic, pushbroom, and two-slit cameras into one family. Companion: Yu, J., & McMillan, L.

(2004). A framework for multiperspective rendering. In *Eurographics Symposium on Rendering (EGSR)*.

- Román, A., Garg, G., & Levoy, M. (2004). Interactive design of multi-perspective images for visualizing urban landscapes. In *Proceedings of IEEE Visualization (VIS 2004)*, pp. 537–544. — the pictorial (non-measured) urban use: seamless pictures serially blended from cross-slit images of broad, roughly planar scenes, with a horizontal path-slit and a user-chosen vertical slit; glide's subject class reached by the picture road rather than the measured one.
- Yang, W., Lin, H., Kang, S. B., & Yu, J. (2015). Resolving scale ambiguity via XSlit aspect ratio analysis. *Proceedings of IEEE ICCV 2015* (preprint arXiv:1506.04338). — the closed-form depth-dependent-aspect-ratio (DDAR) law; states both the perspective aspect-ratio invariance and its cross-slit breaking, with recoverable depth range, sensitivity, and error characterized.

**Context for the construction-versus-computation thesis:**

- Standard treatments of descriptive/synthetic geometry (Monge, Poncelet) and of the double numbers / Minkowski plane; and the historical thread in *Glide Projection* pp. 19–23 (Viola-Zanini 1629; Pozzo 1693; John White, *The Birth and Rebirth of Pictorial Space*, 1957; Klein and Hilbert on the logical basis for multiple geometries).